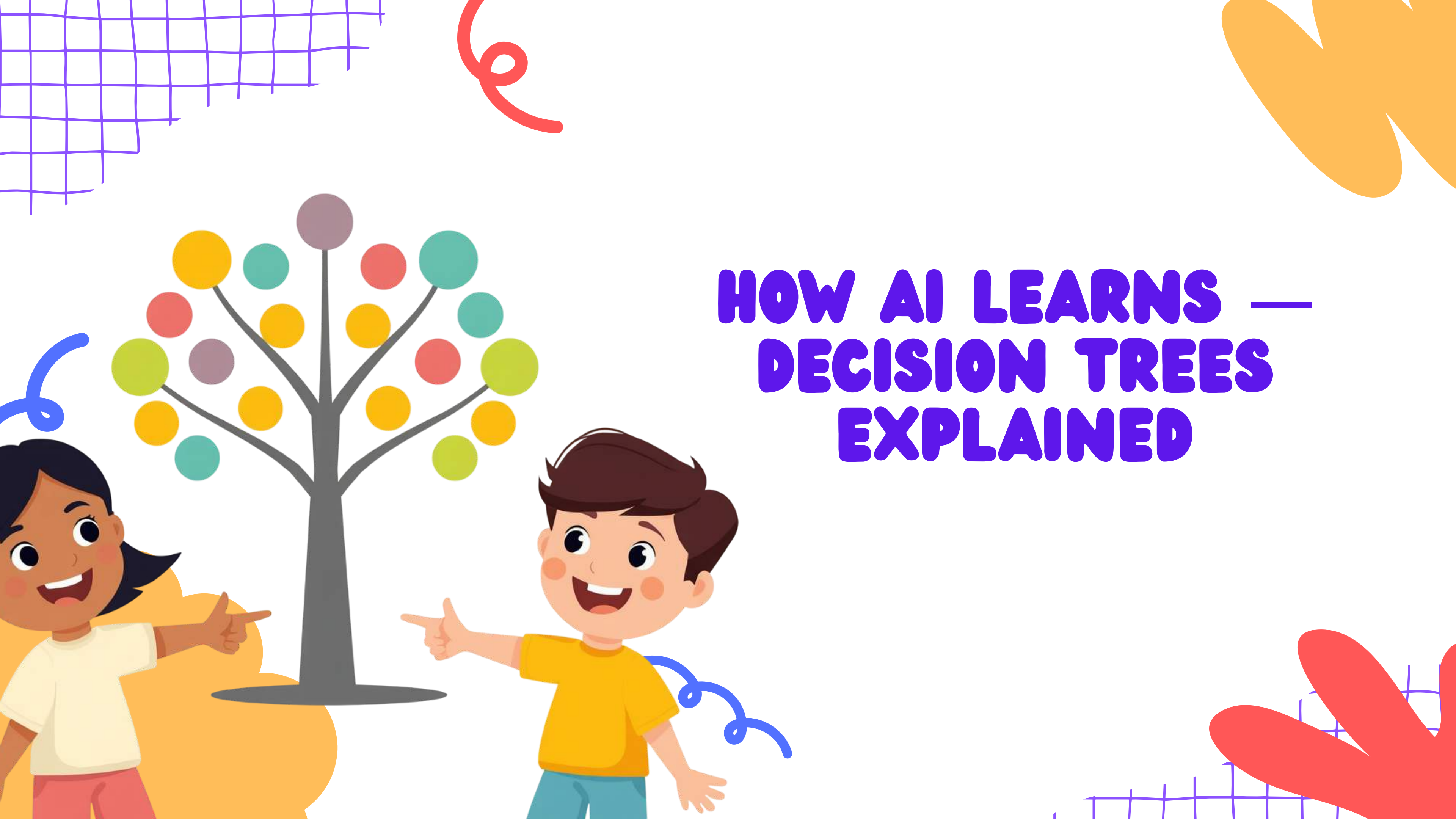


HOW AI LEARNS — DECISION TREES EXPLAINED



WHAT IS AI?

AI means machines that can learn and make smart choices, just like people!

Learn, Smart, Machines — the three big ideas of AI!

AI is all around us — in games, apps, and even your school tablet!



HOW DOES AI LEARN?

AI learns by looking at lots of information and finding patterns. Just like how you look at many pictures to learn what things are, AI studies thousands of examples to understand the world around it!

Like how you learn to tell apples and bananas apart!



A vibrant purple background features a large, stylized white swirl. In the center, a green tree with a brown trunk is depicted. The tree's branches are adorned with circular nodes containing the words 'Yes', 'No', and 'Ia'. Two children, a boy in a light blue jacket and a girl in a yellow dress, stand at the base of the tree, pointing upwards. The scene is decorated with various abstract elements: a white grid pattern in the top left and bottom right, orange and green wavy shapes on the left, and a large orange shape in the top right. A small orange butterfly is also visible near the top center.

A decision tree helps AI make choices, like a game of questions and answers. At every step, it asks a question and follows the "Yes" or "No" path!

 **Every question leads
you closer to the
answer!**



WHAT IS A DECISION TREE?

A decision tree has three important parts. Let's learn what each one does!

Node: A question to ask

Branch: The answer to the question

Leaf: The final choice or result





EXAMPLE: CHOOSING A FRUIT

Let's pick a fruit! Is it red? Yes or No? Then sweet or sour? Follow the questions to find your fruit!

Is it red? Yes → could
be an apple!

Is it sweet? Yes →
apple or mango!

Is it yellow? Yes →
banana or mango!

Is it sour? Yes →
orange or lime!

HOW AI LEARNS THE TREE

AI looks at many examples and finds the best questions to ask. The more examples it sees, the smarter it gets!

Look at data

Ask a question

Split data by
answers

Repeat until done!

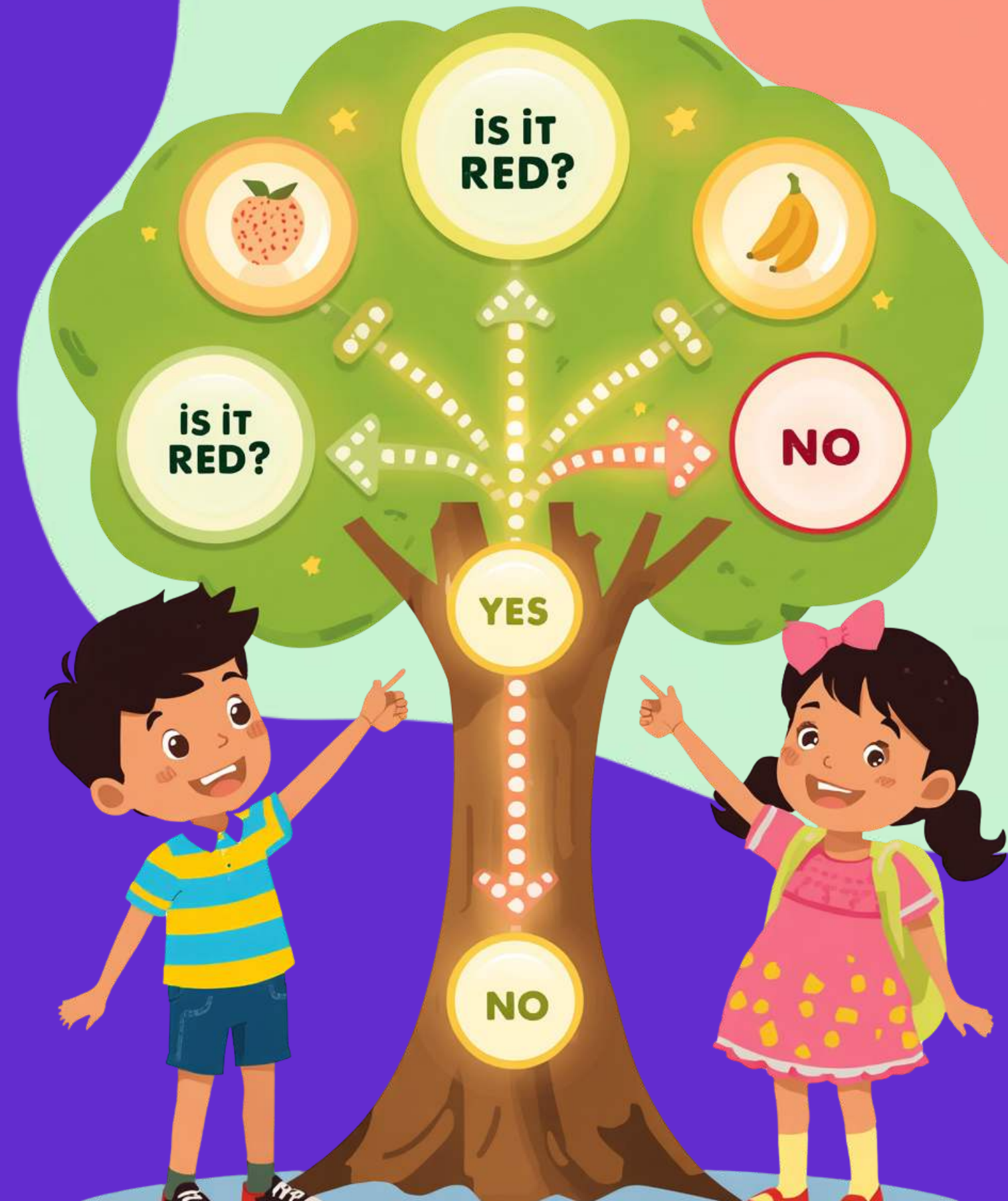


DECISION TREE IN ACTION

Let's see how the AI decides what fruit it is!
The AI starts at the top and asks questions
step by step:

- Is it red? → Yes or No
- Is it sweet? → Yes or No
- Follow the arrows to find the answer!

**Follow the arrows and
guess the fruit with the
AI!**



WHY USE DECISION TREES?

3 Great Reasons!

Decision trees are one of the best tools AI uses to make choices. Here is why they are so amazing!

- ☀ Easy to understand — anyone can follow the questions!
- 📊 Works with lots of data — handles many examples at once.
- ⚡ Helps AI make fast decisions — gets the right answer quickly!



DECISION TREE VS. GUESSING

Let's see the difference between guessing randomly and using a decision tree to make a smart choice!



Guessing: No questions, just a random choice!



Decision Tree: Asks smart questions to choose right!



WHERE DO WE FIND DECISION TREES?

Real-Life Uses of AI!

AI uses decision trees in our everyday lives to help make smart and helpful choices all around us!

-  Helping doctors find illnesses and suggest treatments.
-  Recommending movies or games you might enjoy!
-  Sorting emails into the right folders automatically.



LET'S BUILD OUR OWN TREE!

Your Turn!

Ask yes/no questions to sort your toys or snacks!
Start at the top and branch out with every answer.

"Is it big?" → Yes / No

"Is it soft?" → Yes / No



Try it yourself — draw your own tree and fill in the questions!



QUICK QUIZ!

Test what you know about decision trees! Think carefully and pick the best answer.

1. What is a "leaf" in a decision tree? A) A question B) A final answer C) A branch

1. What does a "node" do in a decision tree? A) Gives the result B) Asks a question C) Ends the tree

1. How does AI learn a decision tree? A) By guessing B) By looking at lots of examples C) By asking a teacher

WHAT WE LEARNED TODAY

Great job exploring AI and decision trees!
Here is a quick recap of everything we discovered together today.

AI learns by asking questions
and finding patterns in data.

Decision trees help AI decide —
with nodes, branches, and leaves!

We can build our own decision
trees to sort toys or snacks!



THANK YOU FOR LEARNING WITH US!

You did an amazing job learning about AI and decision trees today! Remember, every big question starts with a simple "Yes" or "No." Keep exploring, keep asking, and keep learning!

**KEEP ASKING
QUESTIONS AND
EXPLORING AI!**





WANT TO LEARN MORE?

Resources for Teachers & Parents

Ask your teacher or explore kid-friendly AI games online! Books, videos, and fun activities are all around you. Keep asking questions and keep exploring!

Designed by Monu